

Fugacity 2026

ML Hackathon

# The Chemical Engineering Problem

BEYOND BLACK BOX ML: RECOVERING GOVERNING EQUATIONS TO PREDICT  
SERIES PARALLEL REACTOR YIELD

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# Problem & Industrial Relevance

## The Computational Bottleneck

Modern chemical facilities rely on complex computational fluid dynamics and boundary value problem (BVP) simulations. While rigorous, these are too computationally expensive and slow for real time plant optimization.

## System Complexity

The kinetic and thermodynamic behaviors of the reactions create complex, non linear trade offs between flow rates, inlet concentrations, and thermal profiles.

## The Physical System

The process utilizes a continuous flow, non isothermal reactor involving a highly sensitive series parallel reaction network

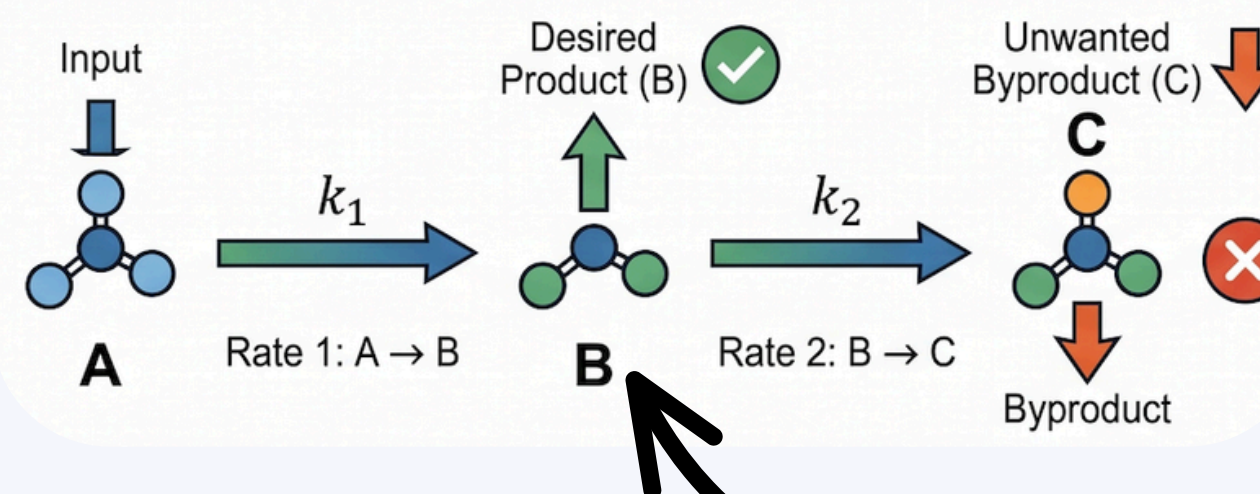
## Industrial Impact

Bridging chemical engineering and computer science to give process engineers a tool for instant, real time operating setpoint adjustments

Dataset : 150 train / 50 test

Metric : RMSE

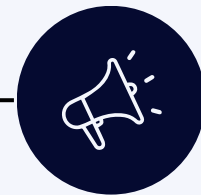
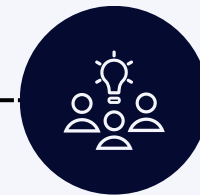
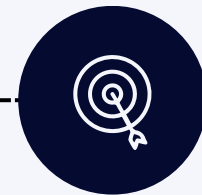
## Non-Isothermal Series Reaction Network ( $A \rightarrow B \rightarrow C$ )



## The Objective

Build a predictive data driven surrogate model to instantly and accurately forecast the overall yield percentage of Product B at the reactor exit.

# Chemical & Process Understanding



## Kinetic Tradeoff

$$E_2 \gg E_1$$

The side reaction ( $B \rightarrow C$ ) is significantly more temperaturesensitive than the desired reaction ( $A \rightarrow B$ ). Consequently, raising the temperature destroys Product B faster than it synthesizes it.

## Thermally Self Destabilizing Flow

### Goal Setting

- The primary reaction  $A \rightarrow B$  is endothermic ( $\beta_1 < 0$ ), causing the feed to selfcool upon entry.
- The secondary reaction  $B \rightarrow C$  is exothermic ( $\beta_2 > 0$ ), releasing heat downstream exactly where Product B concentration peaks and is most vulnerable to degradation.

## The Jacket Only Fallacy

### Strategy Design

Models utilizing only a jacket energy balance fail because they are forced to reproduce a sharp thermal switch with a single Arrhenius term. This results in a physically impossible activation energy of  $E_2 \approx 634 \text{ kJ/mol}$ . Incorporating both reaction enthalpies successfully resolves this.

## Residence Time Dynamics

### Preparation

The yield versus  $\tau$  (where  $\tau = L/Q$ ) exhibits an interior maximum. This optimum shifts to shorter residence times ( $\tau$ ) as the mixture temperature ( $T_{\text{mix}}$ ) rises.

### First Order Kinetics Supported

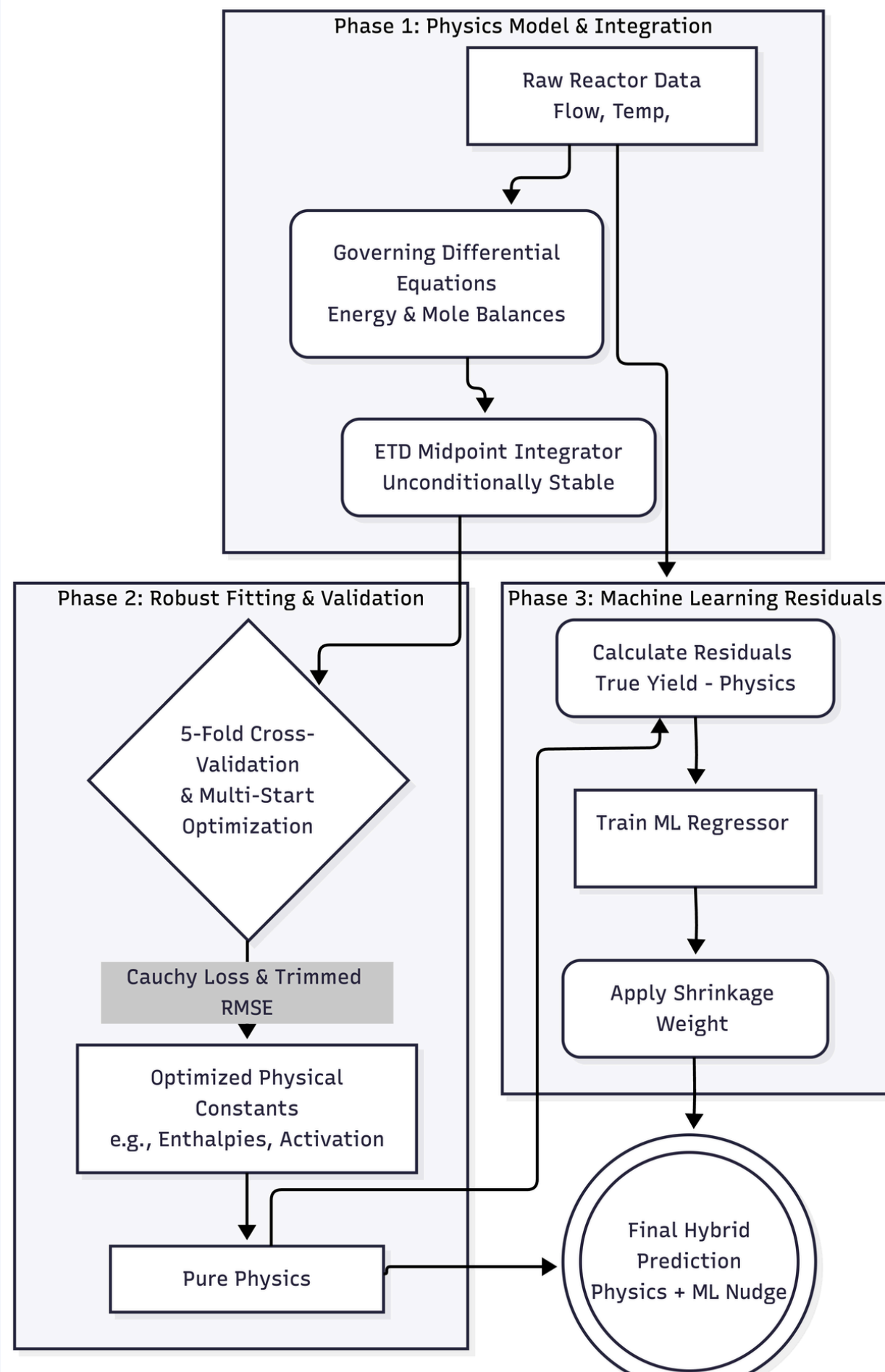
Yield is independent of inlet concentration ( $r = 0.009$ ), consistent with first order. Freeing the order worsens CV, so we keep  $n = 1$ .

## Validating "ZeroYield" Data

### Execution Stage

The 37 exact zeroyield rows are not sensor failures or missing data. They represent a physical selectivity cliff past 460 K where Product B is completely consumed into Byproduct C.

# ML Methodology & Approach



## GOVERNING EQUATIONS

Recovered the **exact differential equations** for energy and mole balances instead of relying on black box ML

01

## STABLE INTEGRATION

Deployed an **unconditionally stable ETD midpoint integrator** to accurately handle sharp thermal switches in the reactor.

02

## ROBUST FITTING

Inferred data contamination to justify using a **Cauchy loss function** for highly robust parameter optimization

03

## CROSS VALIDATION

Executed strict **5-fold crossvalidation** alongside 24 wide multistart restarts to secure the true global optimum.

04

## ML RESIDUALS

Trained an **ExtraTrees regressor** solely on physics residuals, using a **deliberate shrinkage** weight to prevent overfitting

05

## HYBRID PREDICTION

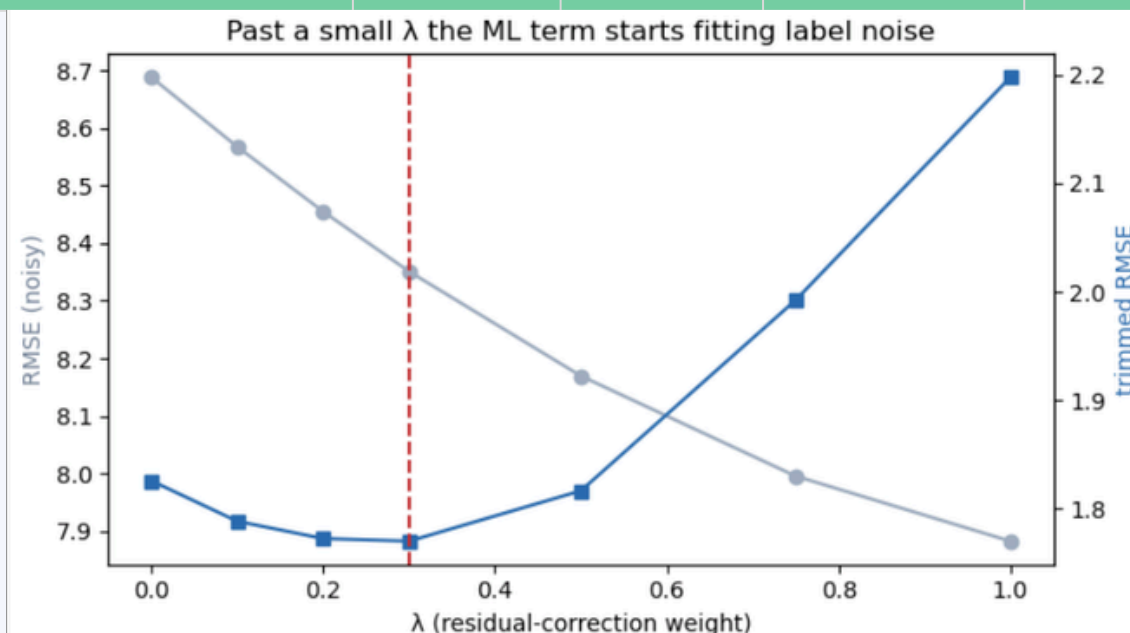
Combined the dominant physical model with the **minor ML correction** to generate the highly accurate final prediction.

06



# Results & Model Performance

| Model                                     | Params | CV RMSE     | Trimmed RMSE | MedAE       |
|-------------------------------------------|--------|-------------|--------------|-------------|
| Pure ML<br>(ExtraTrees, 8 feats)          | -      | 18.09       | 12.39        | 9.05        |
| Jacketonly energy balance                 | 5      | 10.76       | 3.4          | 1.22        |
| + $\Delta H$ of $A \rightarrow B$         | 6      | 8.84        | 2.97         | 0.94        |
| + $\Delta H$ of $B \rightarrow C$         | 7      | 8.32        | 1.83         | 0.84        |
| + shrunk ML residual<br>( $\lambda=0.3$ ) | 7      | <b>7.98</b> | <b>1.77</b>  | <b>0.72</b> |



01

## Physics Outperforms ML

Pure ExtraTrees scores 18.09 RMSE; the 5parameter physics model, with no ML, scores 10.76.



02

## The Ablation Ladder

Every performance gain came from adding a physical term, never from adding model capacity.



03

## Corrected Kinetics

Modelling both reaction enthalpies dropped apparent  $E_2$  from 634 to 242 kJ/mol impossible to defensible.



04

## Why $\lambda = 0.3$

The 5 parameter physics beats ML by 3.6 $\times$  on trimmed RMSE. ML enters only as a shrunk correction.



05

## Opposing thermal effects

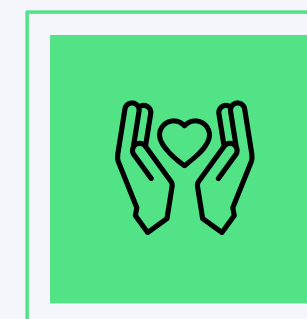
$\beta_1 = -40.8$  K,  $\beta_2 = +37.4$  K: the feed selfcools, then the side reaction reheats downstream.



06

## Leak free validation

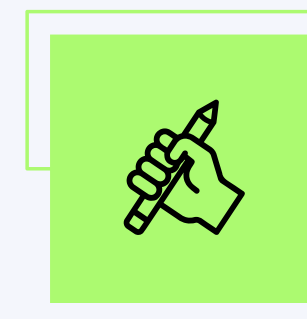
5-fold CV repeated over seeds, with the physics refit inside every fold.



07

## Physically consistent

Reproduces the interior optimum in  $\tau$  shifting shorter with temperature, and the 37 zeroyield runs.



08

## Stated limitations

$\lambda$  is weakly identified and tuned on the scoring CV;  $E_2$  stays apparent, not mechanistic.

Leaderboard RMSE: 11.2

# Failure Analysis & Limitations

01

## Catalyst deactivation

Activity  $k_1$  falls over time; the model assumes it constant and overpredicts yield.

02

## Jacket fouling

Heat transfer  $h$  degrades, shifting the thermal switch and moving the optimal residence time.

03

## Feed or flow change

Inerts, cofeeds, or laminar flow break the firstorder and idealplugflow assumptions.

04

## Operating near the cliff

Above  $\sim 460$  K yield collapses steeply; small temperature errors become large prediction errors

05

## Extrapolation limits

Physics stays valid outside the training box; the tree residual term flattens and misleads.

06

## Weakly identified parameters

Reaction order's interval is  $[0.87, 1.91]$ ; a residual correlation with  $Q \cdot C_0$  survives at  $r \approx 0.22$ .

07

## Diagnosing the source

Refit online and see which constant moves:  $h$  means fouling,  $k_1$  means deactivation, offset means sensors.

08

## Unmodelled parallel path

The brief calls the network series-parallel; we modelled  $A \rightarrow B \rightarrow C$ . A genuine  $A \rightarrow C$  route would show as a systematic offset.

# Thank You!!

Team Name : Outliers